The Law of Graphic Equalisation — Born Recovery

An Axiom-by-Axiom Note

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DRAFT — draft-2026-09-06.1r6 (6 September 2026). This document is under active revision and is circulated for comment, not as a finished result. It is a companion to *The Law of Graphic Equalisation — Theory* (referred to below as Graphic Equalisation) and to *The Law of Graphic Equalisation — Physics Grounding* (the physics companion), and inherits those papers' definitions, epistemic boundary, and level-of-claims; *The Law of Narrative Momentum* (the momentum paper) supplies the Narrative and the first law. It is an *attempt*, rated step by step, and not a peer-reviewed result. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

This note maps the Graphic Equalisation substrate to operational reconstructions of quantum theory. Weighted realisations motivate the classical probabilistic core. The quantum-selecting axioms remain obligations: free continuation does not establish Hardy's continuous transitivity between pure states, and restriction to a larger frame does not establish purification unique up to reversible transformations. Conditional Born recovery requires complex linear structure, composition and probability assumptions. The rated mapping records these gaps rather than treating them as native derivations.

1 The substrate, in one paragraph

Seven notions carry the mapping below. Each is defined in one of the three papers named above; this section fixes the references and adds nothing. A *frame* is a subgraph that runs (Graphic Equalisation's Definition 6.2), and frames compose by *nesting*, a graph being a subset of its container (its Proposition 6.6). A frame's state is its *Narrative*, the standing structure it carries forward (the momentum paper's Definition 2.2), and where a frame instantiates a direction — optional in general, present in this setting — that Narrative has a magnitude and a direction (the momentum paper's Definition 4.1); the state proper is the *internal* Narrative, as against what another frame reads off it (its Definition 5.3). A *closure cycle* is a recurrence a frame may provisionally treat as closed, the simplest operational clock model rather than a condition every realisation must satisfy (Graphic Equalisation's Proposition 13.7); its execution may be locally closure-driven, externally clocked, or superframe-scheduled. *Phase* is position on that modelled recurrence rather than proof of a signed substrate time (its Remark 13.8). A *realisation* is a collapse, a local write that picks one outcome from the admissible set and carries a weight (Graphic Equalisation's Definition 4.1), and realisations are strictly ordered, the order carried by the rules rather than by a clock (its Proposition 11.2). *Free continuation* is what a frame does when no rule fires, the information-preserving default (the momentum paper's Definition 5.2); *free* means information-preserving and not absent, since evaluating a rule is itself

a write (Graphic Equalisation's Proposition 7.16), and all the mapping below uses is that free continuation is *injective*, hence reversible. Nothing is privileged: there is no god's-eye frame, and therefore no privileged phase origin and no privileged measurement basis (its Axiom -1). *Entanglement* is a superframe dimension: a subframe's local state is mixed because the purity is a dimension of the larger frame (the physics companion's Proposition 6.1 and its Corollary 6.2).

2 The claim, honestly sized

The note proposes a conditional recovery of the Born rule. It can invoke reconstruction results only after an operational model meeting their full hypotheses has been constructed; that mapping remains incomplete.

Remark 2.1 (Recovery is the honest verb, and the right one)*.* Derivation per se would compel Born from primitives. The framework forces almost nothing, and that is the point of a graph substrate: what survives every attempted removal is a container relation across which rules run (Graphic Equalisation's Remark 3.2), and essentially everything else is contingent rule-content, as with the number of spatial dimensions (the physics companion's Section 15) or the gauge group (its Remark 11.3). A contingent law can only be recovered given its rule-content, never derived as compelled. Some steps below are forced *given the premises* — the squared modulus under the separately assumed noncontextual additive probability measure and Hilbert structure — while the premises themselves are admitted and contingent: complex, linear, self-similar.

The distinctively quantum ingredients are proposed identifications whose premises still need to be constructed; the cited reconstruction theorems supply the uniqueness (Gleason, 1957; Hardy, 2001; Masanes and Müller, 2011; Chiribella et al., 2011). The only real claim is the *arrangement* — each component is individually known. That sizing is not modesty about the result but a statement of what kind of result it is, and it governs every rating below.

3 Level 0: the substrate is an operational-probabilistic theory

Before any reconstruction axiom applies, the substrate must be an operational-probabilistic theory (OPT): states in a convex set, effects, transformations, and a composition rule. That framework is a *precondition* for either axiom set, so the mapping has two levels, and this is the lower one.

The key that makes it tractable is seeing what the OPT convex structure *is*. The convexity of the state space and the affinity of effects are the *classical total-probability backbone* — total probability over ordered branchings — distinct from, and prior to, the quantum superposition linearity that Section 4 worries about. Every probabilistic theory has this backbone. A weighted grounding can have it; normalised weights and mixture operations are additional premises, not consequences of deterministic realisation alone.

Remark 3.1 (A candidate operational-probabilistic grounding)*.* If normalised realisation weights and probabilistic mixing are added, ordered realisations can support convex states and affine effects via total probability. The substrate's deterministic transition definitions alone do not supply these weights. Operational equivalence, identities, closure under transformations, and consistent sequential and parallel composition must also be constructed.

Ingredient by ingredient, with the substrate structure that supplies each. Four ratings are used here and in Section 4: *solid*, the substrate structure is the requirement's statement; *earned*, derived here from a substrate principle rather than read off one; *plausible*, a commitment of the right kind, with statement-match not shown; *contingent*, rule-content, admitted and self-located rather than supplied.

OPT requirement	Supplied by	Rating
Systems	Frame types: a subgraph together with an outcome set.	solid
States are well-defined equivalence classes	Given an operational state that includes all causally relevant memory, statistics depend only on the current Narrative — history is optional and present structure acts, so there is no preparation-memory and operational equivalence must be checked.	plausible
Convex state space	Mixing two preparations by a weighted coin-realisation: realise a λ-weighted coin, prepare P on one branch and Q on the other.	solid
Affine effects	Total probability over the coin–prepare–measure branching, $e(\lambda P + (1-\lambda)Q) = \lambda e(P) + (1-\lambda)e(Q)$.	solid
Transformations (monoid, with reversibles)	Rules compose in order; identity, closure and allowed inverses need explicit construction. A changing free continuation is not necessarily the identity.	plausible
Composition	Subframe nesting; a joint preparation lives over the product outcome set. Existence is Level 0; the complex-specific tomography is the contingent layer of Section 5. Parallel composition still needs its axioms checked.	plausible
Probabilities	Realisation weights in $[0, 1]$, normalised over a complete measurement.	solid

Remark 3.2 (The affinity is not the superposition linearity). The affinity of effects is the law of total probability applied to ordered realisations, and nothing about it is quantum. It is worth marking because the two linearities are easy to run together, and the argument would be circular if they were: a weighted grounding supplies the classical backbone under its stated probability premises, and Section 4 must earn the superposition linearity separately, on other grounds.

The remaining prerequisites are substantive: a finite tomographic dimension, in the sense that a state is fixed by finitely many fiducial effects — plausible for finite frames by the aperture, unproven in general; topological closure and compactness of the convex state space; the full composition axioms, associativity of sequential and parallel composition and the interchange law, which should follow from the ordering and nesting of realisations but each need checking; and the no-restriction and effect-realisability niceties. Convexity and affinity likewise depend on the specified mixing and probability operations; they are carried forward in Section 7.

4 The distinctively-quantum axioms, mapped

Two independent reconstructions are used. Hardy (2001) gives five axioms, of which classical probability theory satisfies the first four and fails the fifth. Chiribella et al. (2011) — Chiribella, D'Ariano and Perinotti, CDP below — give six principles that single out quantum theory over the real, the quaternionic and the classical. They have *different* most-quantum axioms, which is what makes the pair worth running rather than either alone.

Remark 4.1 (The two quantum-selecting axioms remain obligations). Hardy's axiom requires a continuous reversible path between any two pure states (Hardy, 2001). Global $U(1)$ phase acts trivially on rays and does not connect distinct pure states. Free continuation supplies no such transitive action. CDP purification requires a pure extension of every state and uniqueness up to a reversible transformation on the purifying system (Chiribella et al., 2011). The superframe reading suggests restriction of a pure joint state, but proves neither general existence, operational purity, nor the uniqueness clause. Both correspondences are plausible targets.

The two crux axioms are distinct targets, and neither is established by the substrate mapping. The table records that status:

Axiom	Substrate structure	Rating
Continuity of reversible transformations (Hardy; classical theory fails exactly this)	Free continuation: information-preserving, hence reversible, with continuous cycle-phase, but no transitive action on pure states established.	plausible
Purification (CDP; fails classically)	Entanglement as a superframe dimension: a mixed subframe state is the restriction of a pure superframe state; general existence and uniqueness are unproved.	plausible
Phase- and context-independence	No privileged phase origin and no privileged frame: probability is invariant under global phase and the same across measurement bases; projector noncontextuality needs a separate premise.	plausible
Probability	Weighted realisations; frequencies over an ensemble of collapses from one possibility structure.	plausible
Simplicity and ideal compression	Minimum description length as a native commitment (Graphic Equalisation's Remark 2.7) (Rissanen, 1978).	plausible
Subspaces	Subframe-as-frame: restriction to a subset of distinctions behaves as a smaller frame.	plausible
Perfect distinguishability	Distinctions are first-class; a state that is not completely mixed carries one.	plausible
Pure conditioning	Collapse of a pure Narrative yields a definite realised outcome.	plausible
Local tomography (Hardy's composite axiom; CDP's local distinguishability)	Subframe composition over the product outcome set — *contingent*, and the subject of Section 5.	contingent

The plausible ratings are genre-matches rather than statement-matches: the substrate has a commitment of the right kind, and that the specific axiom is the one the substrate enforces is not shown. Marking them so is the point of the table; a reader who upgrades one of them is disagreeing with a rating, which is the intended form of the disagreement. The table is a correspondence map, not a proof that the operational axioms follow from the substrate: "solid" marks a direct structural match, while "earned", "plausible", and "contingent" mark respectively an argument still carrying a premise, a right-shaped identification, and admitted rule-content. Only the cited reconstruction theorems establish their own axiomatic consequences once their premises are granted.

The squared modulus is selected among the natural candidates by phase-averaged additivity, and made unique by Gleason, in the physics companion (its Remark 7.1 and the argument following it); this note uses the result. What the framework supplies to Gleason (Gleason, 1957) are his premises — a vector-valued state, from the linearity step, and a context-independent measure, from an additional noncontextuality assumption — and Gleason supplies what the framework does not.

Remark 4.2 (Two supporting identifications). The proposed inner-product interpretation is a cross-frame projection. Constructing its algebraic properties remains necessary; the metaphor does not supply an inner product. *Reversibility* is recoverability from an injective free continuation. Unitarity additionally requires linearity and inner-product preservation; destroying no distinction alone is insufficient — an information-preserving write rather than the absence of one, which is the right shape, since unitary evolution plainly does change the state and is reversible for exactly that reason. Both are identifications rather than derivations, and the rigorous step from "uncoupled" to full additivity and norm-preservation is the standing residual of Section 7.

5 Complex over real: contingent, not forced

Local tomography is the axiom that selects complex Hilbert space over real, and it is where a forcing argument would have to sit. Writing K for Hardy's fiducial degrees of freedom including normalisation, for two two-state systems:

Theory	K **single**	K **composite**	$K_A \cdot K_B$	**Local tomography**
Classical	2	4	4	holds (but fails continuity)
Real (rebit)	3	10	9	**fails** ($10 > 9$)
Complex (qubit)	4	16	16	**holds**

A closure-phase frame is *consistent with* complex amplitudes at one system — the Bloch sphere, $K = 4$, rather than the real disk's $K = 3$ — though a phase-less frame is equally constructible and the substrate requires neither. Assuming the usual complex tensor-product state space gives $K_{AB} = 16$ and local tomography. Being a superframe alone does not establish that composition rule.

Remark 5.1 (Complex versus real composition is an open substrate mapping). Standard complex two-qubit composition has $K_{AB} = 16$, whereas real two-rebit composition has $K_{AB} = 10$. These are candidate physical groundings; this note does not construct either as an admissible substrate configuration. Whether both are available, and which selection principle applies, remains open.

Remark 5.2 (Contingency needs a construction). A substrate need not force all physical rule-content. To claim that it admits both real and complex quantum theories, however, requires explicit admissible models of both, including their composition rules. The fiducial counts of the standard theories do not provide those constructions or a selection principle.

If the full mapping is established, the Born rule and Hilbert structure follow via the cited reconstructions — which is how the framework treats all physics: shapes follow from contingent rule-content, the rule-content itself never forced. Of the post-2011 reconstructions, Masanes and Müller (2011) obtain finite-dimensional quantum theory from five physical requirements on states and transformations, and theirs is the formulation this note would most like tested against, since those requirements are stated on single systems plus a composition rule — exactly the seam where the substrate's answer turns contingent rather than forced. Independent support for reading the selection as contingent comes from Höhn (2017) and Höhn and Wever (2017), whose rules on an observer's acquisition of information yield the correlation structure of qubits and rebits alike and so do not by themselves select complex over real. The rest of the reconstruction landscape is the physics companion's (its Section 17).

6 Relation to the clock programme

The Born mapping supplies no relativistic clock law or equivalence-principle calculation. The reconciliation companion's Proposition 3.2 states the boundary on the physics companion's budget reading. Conditional measurement reconstruction also requires its own assumptions (Masanes et al., 2019; Schlosshauer and Fine, 2005).

7 Residuals

What is open, unaddressed, and named:

1. **Level 0 hygiene.** Whether "present structure acts" genuinely delivers operational equivalence — states as equivalence classes of preparations with identical statistics — and whether the full OPT composition axioms (associativity of sequential and parallel composition, the interchange

law) hold under the ordering and nesting of realisations. With them, a finite tomographic dimension, topological closure and compactness of the state space, and no-restriction and effect-realisability. These are hygiene rather than core, and plausible for finite frames; none is proven here.

2. **The linearity step.** The rigorous passage from "uncoupled free continuation" to full additivity and norm-preservation. The mechanism motivates linearity; it does not prove it, and the linearity of the imposed supergraph map is not handed over for free. This is the one posit the recovery rests on.

3. **The squaring argument.** Whether the phase-average selection is tight enough on its own, or whether the full Gleason content is needed to reach uniqueness. As the physics companion states it (its Remark 7.1 and the argument following it) the phase average selects rather than proves, and Gleason is cited for the rest.

4. **Composition.** Under the natural reading the composite is local-tomographic and complex. Whether that reading is forced, permitted, or wrong is not settled here. The formulation of Masanes and Müller (2011) is the one this seam should be tested against, its requirements being stated on single systems plus a composition rule; that is exactly where the substrate's answer turns contingent rather than forced, and where the mapping is least confident that it is doing honest work.

5. **The plausible ratings.** Probability, simplicity and compression, subspaces, distinguishability and pure conditioning are genre-matches to native commitments, not statement-matches. Each would need tightening before the cited theorems could be applied without qualification.

8 Failure conditions of the mapping

Each commitment below is marked [SHAPE] or [DETAIL]. A [DETAIL] failing is a revision: the account survives with that mechanism replaced. A [SHAPE] failing costs *range* rather than a mechanism, because what fails is something the substrate is claimed to require or to permit — so the account does not hold where it claimed to, which is the more serious of the two and is still a boundary rather than an annihilation.

1. [SHAPE] **The operational-probabilistic instantiation.** Fails if the substrate does not instantiate an OPT — if convex states, affine effects, a transformation monoid with reversibles, composition and normalised probabilities cannot all be had from weighted ordered realisations and the history-optional first law. Everything below is conditional on this, so it fails with it.

2. [DETAIL] **The Level-0 hygiene conditions.** Fails if the conditions of Section 7 — a finite tomographic dimension, topological closure and compactness of the state space, the full composition axioms — cannot be met for the frames this note is about. Each is a mechanism the note could lose and replace without losing the instantiation above.

3. [DETAIL] **Linearity from free continuation.** Fails if the step from uncoupled independent continuation to full additivity and norm-preservation cannot be made rigorous — in which case the vector structure is a posit rather than a recovery, and Gleason's first premise is imported rather than supplied.

4. [SHAPE] **The axiom mapping.** Fails if it does not go through for a named axiom: in particular if free continuation is not Hardy's continuous reversible transformation between pure states, or if entanglement as a superframe dimension is not CDP's purification. Failure at either of those two is worse than failure elsewhere, since they are the axioms that separate quantum from classical and are what Remark 4.1 claims.

5. [DETAIL] **The squared modulus.** A conditional recovery must meet the chosen theorem's full hypotheses. No privileged phase and linear amplitudes alone are insufficient. Phase-averaged additivity is a separate probability postulate; a Gleason argument additionally requires noncontextuality.

6. [DETAIL] **The composition seam.** Construct a joint state space and consistent composition operations, then test the selected reconstruction's requirements. Standard real and complex fiducial counts are comparison cases, not an exhaustive list of substrate possibilities.

7. [DETAIL] **Complex over real is contingent.** Demonstrating contingency requires admissible constructions with different composition rules. Until both models are supplied, this is an open question rather than an established range claim.

Scope note (not a falsifier). This note is shape-level and not quantitative: it fixes no dimensionless constant, derives no field equation, and offers no theory of quantum gravity. Nor is it a derivation: the claim is the *arrangement*, and no step above is offered as compelling the Born rule from primitives.

References

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